RAJALAKSHMI ENGINEERING COLLEGE

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Y2

Started on	Wednesday, 30 September 2026, 9:36 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:37 AM
Time taken	48 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	**x**	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

#define MAX(a, b) ((a) > (b) ? (a) : (b))

int main() {
    char s1[1005];
    char s2[1005];

    // Read the two strings
    if (scanf("%s", s1) != 1 || scanf("%s", s2) != 1) {
        return 0;
    }

    int m = strlen(s1);
    int n = strlen(s2);

    // Allocate memory for two rows to optimize space complexity
    int *prev = (int *)calloc(n + 1, sizeof(int));
    int *curr = (int *)calloc(n + 1, sizeof(int));

    if (prev == NULL || curr == NULL) {
        return 1;
    }

    // Dynamic Programming transitions
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                curr[j] = 1 + prev[j - 1];
            } else {
                curr[j] = MAX(prev[j], curr[j - 1]);
            }
        }

        // Move current row data to previous row for the next character iteration
        for (int j = 0; j <= n; j++) {
            prev[j] = curr[j];
        }
    }

    // Print the final maximum length
    printf("%d\n", curr[n]);

    // Free memory allocations
    free(prev);
    free(curr);
```

```
49
50      return 0;
51  }
52
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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